Master Control Lead Scoring and Conversion Optimization

Predicting high-value leads to boost sales efficiency

MasterControl

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The Problem

SDRs operate with limited time and lack a data-driven system to prioritize which leads to contact first, resulting in inefficient resource allocation.

Cost Reality

Each sales call costs $50 in SDR time and resources, while each successful conversion generates $6,000 in value.

Revenue Impact

The 7% conversion gap between Mx and Qx translates directly to substantial lost revenue and missed growth opportunities.

Strategic Need

Intelligent lead prioritization is essential to maximize ROI by directing SDR efforts toward highest-probability conversion opportunities.

Project Objective

Predictive Model

Build a machine learning lead scoring model to identify and rank high-probability conversion opportunities using historical CRM data.

Data-Driven Ranking

Generate probability scores for all leads and rank them by conversion conversion likelihood, enabling systematic prioritization.

ROI Maximization

Optimize sales team efficiency and revenue impact by directing SDR outreach to the right leads at the right time.

Dataset Overview & Key Traits

Scale

Approximately 16,800 leads analyzed across Mx and Qx product lines

Imbalance

Severe class imbalance with ~85% non-converted leads requiring specialized modeling techniques

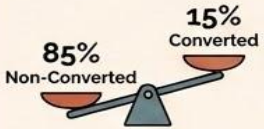
Conversion

Overall conversion rate of 18% across the entire dataset

Performance Gap

Distinct 7% performance gap between Mx (12.7%) and Qx (19.7%) segments

Key Technical Challenges

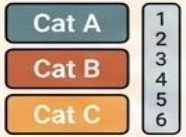


Severe Class Imbalance

- **85% non-converted** leads required **SMOTE**, class weighting, and threshold optimization techniques.

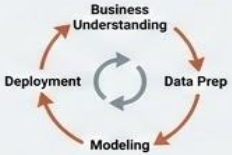
Business Explainability Requirement

- Sales adoption required **SHAP values** and interpretable feature importance for actionable insights.



Categorical Feature Dominance

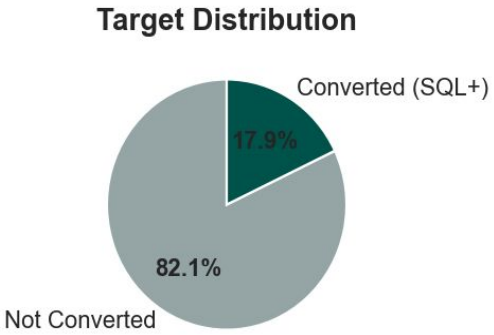
- Limited numerical data necessitated advanced encoding strategies and feature interaction engineering.

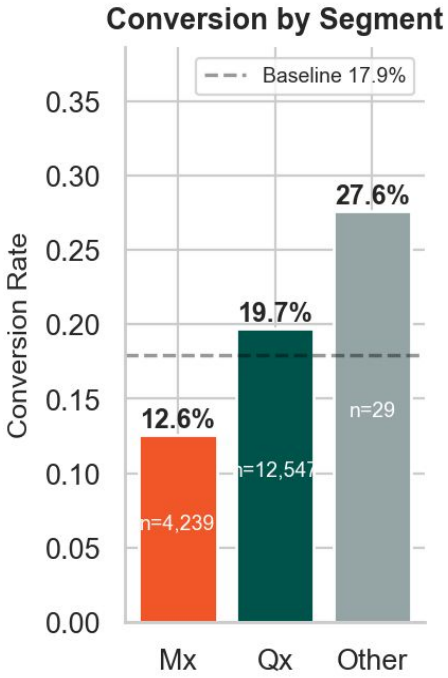


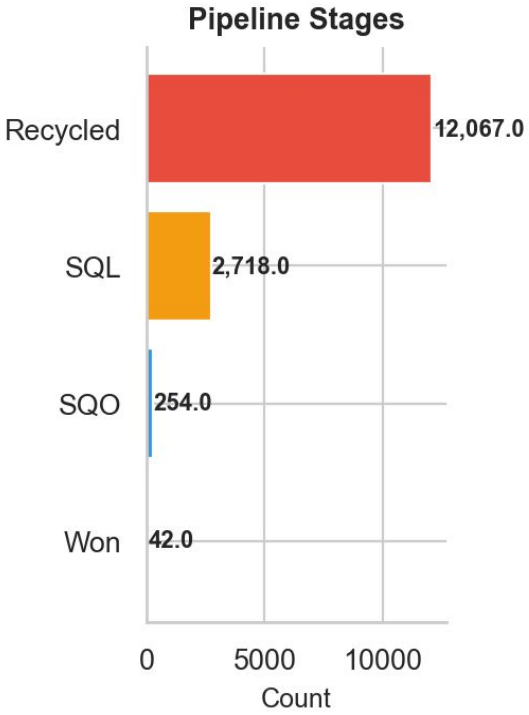
CRISP-DM Framework

- Structured methodology ensured business understanding, data preparation, modeling, and deployment alignment.

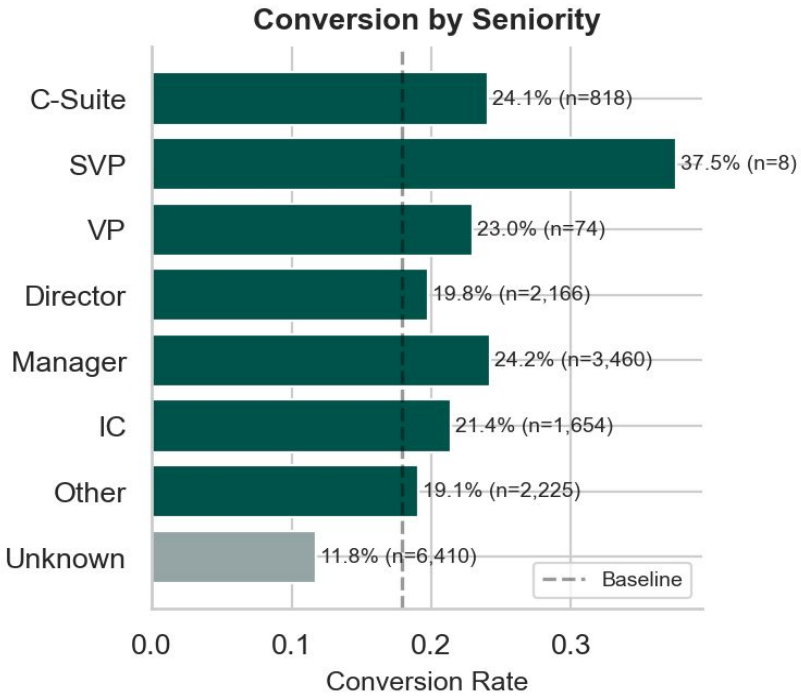
Class Imbalance in the Data

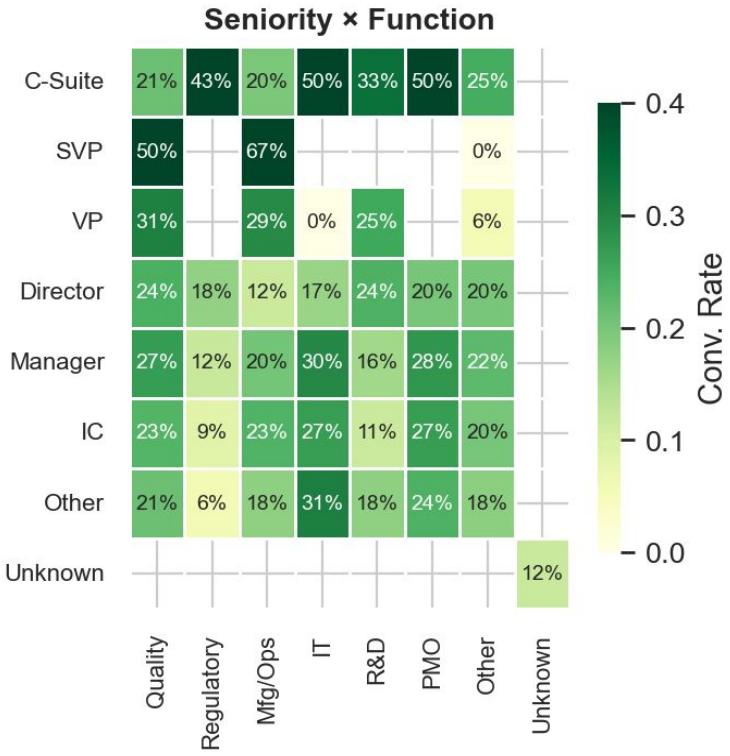






Free-Text Job Title Complexity





Core Feature Engineering Strategy

Intent Strength	Content downloads, demo requests, webinar attendance signal buying readiness
Channel Efficiency	Premium channels (webinars, direct outreach) vs. low-value channels (organic search)
Firmographics	Company size and industry segmentation identify ideal customer profiles
Role Seniority	Decision-making authority indicators (VP, Director, C-suite) predict conversion
Behavioral Recency	Time-based engagement signals capture momentum and buying cycle timing

Key Outcome: Strong separation achieved between converted and non-converted lead cohorts

Advanced Feature Engineering

Buying Committee Activity	Cross-Product Interest	Executive Momentum	Regulatory Urgency Scoring
Multiple stakeholders from same organization signal organizational buying intent	Mx + Qx combined engagement indicates broader solution needs and higher deal value	Seniority × intent interaction captures executive-level buying signals and urgency Industry compliance pressure (Pharma ra, extract decision-making Medical Device) predict quality system needs	Job title embeddings and role classification extract decision-making authority from unstructured text

These advanced features significantly improved model discrimination power

Exploratory Data Analysis: Key Findings

Logistic Regression	Baseline model providing interpretability and coefficient-based feature importance
Random Forest	Non-linear pattern detection with feature interaction discovery
Gradient Boosting Suite	XGBoost, LightGBM for advanced tree-based sequential error correction
CatBoost	Optimized for categorical features with built-in encoding strategies
Ensemble Techniques	Stacking and voting methods combined model strengths for optimal performance

Modeling Approach & Selection

Logistic Regression	Baseline model providing interpretability and coefficient-based feature importance
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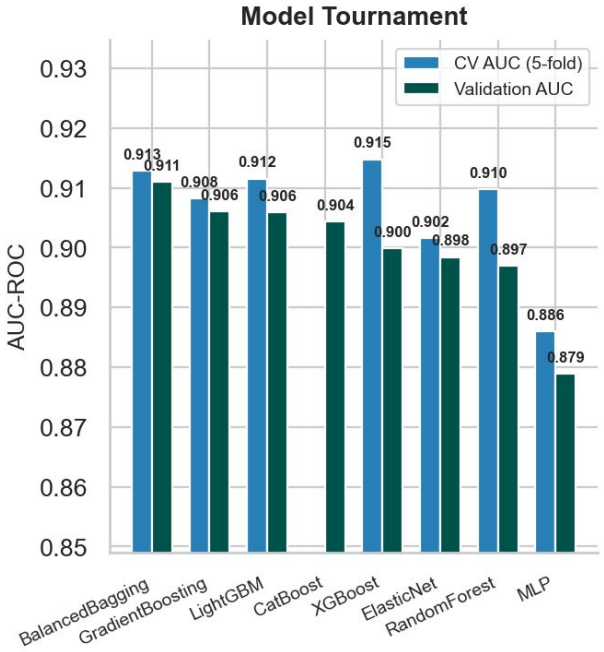
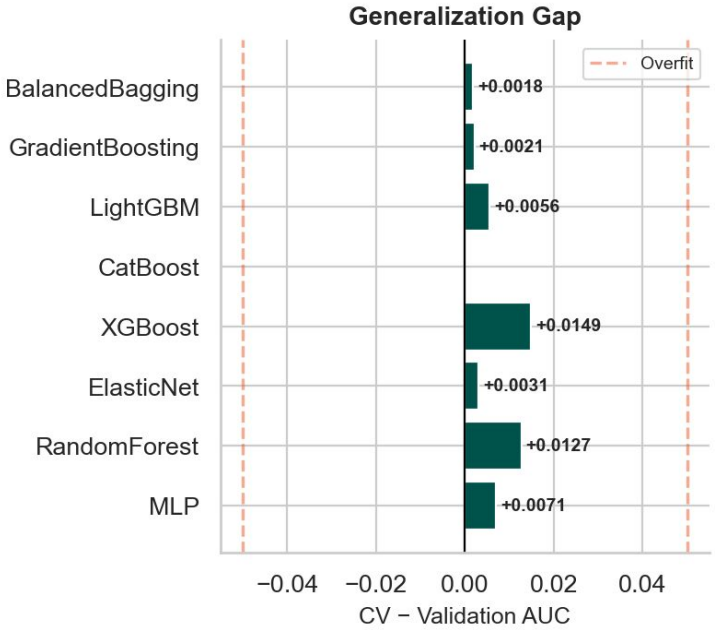
Model Evaluation Strategy

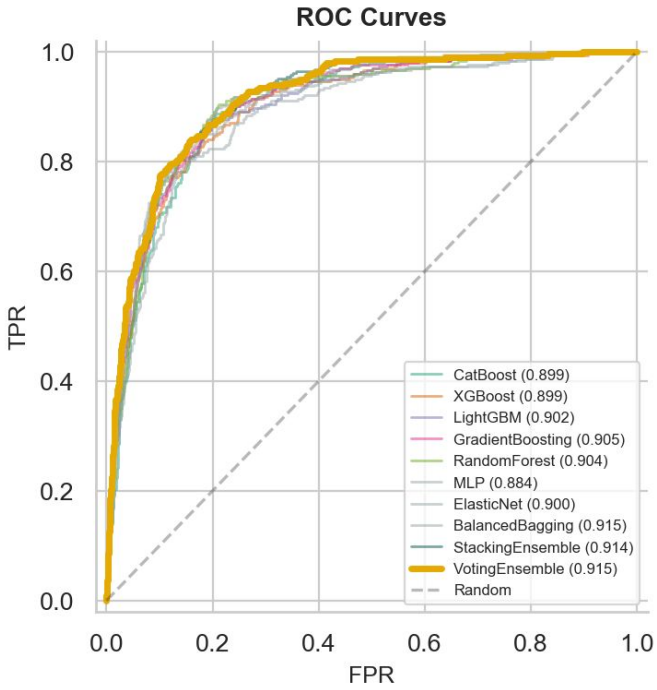
Primary Metric	AUC-ROC measures ranking performance and lead prioritization effectiveness
Metrics	Precision, Recall, F1-score, and calibration analysis ensure balanced performance
Generalize	Rigorous test set validation prevents overfitting and ensures real-world applicability
Business Fit	Prioritized actionable insights over raw accuracy for sales team adoption
Cross-Validation	Stratified k-fold validation ensures robust performance estimation across data splits

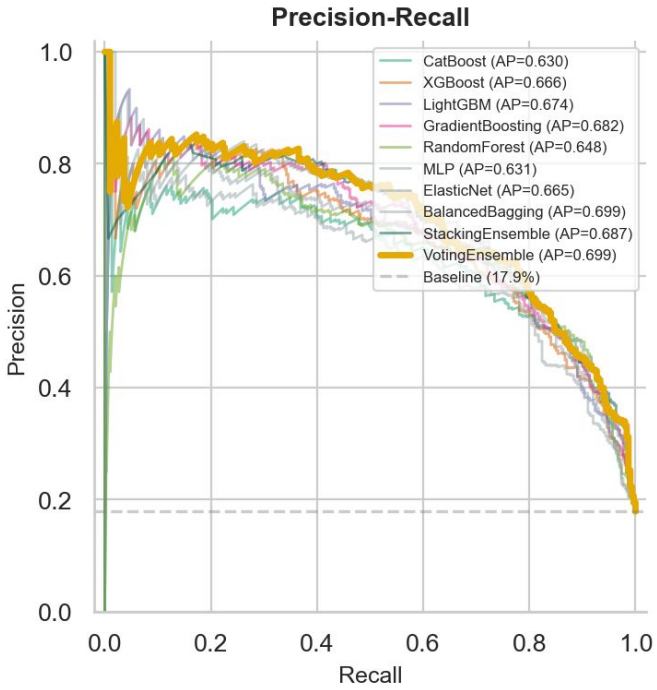
Model Evaluation Strategy Visualization Part 1



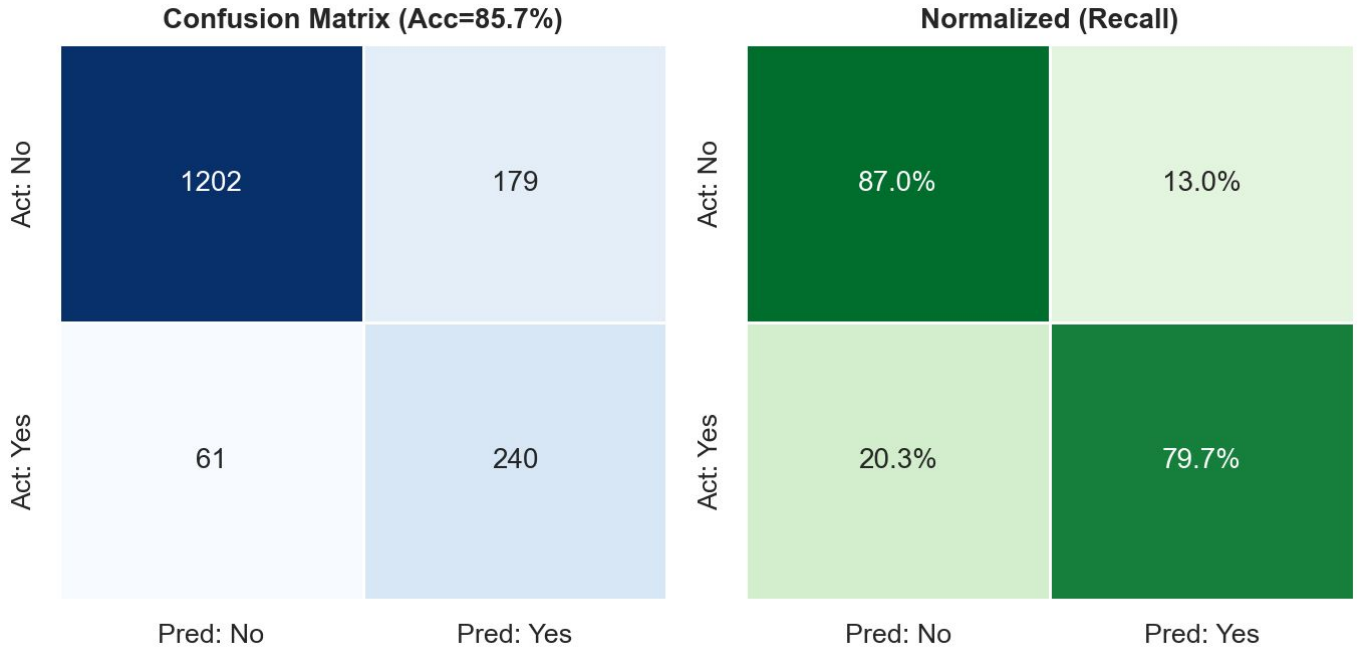


Model Evaluation Strategy Visualization Part 2





Model Evaluation Strategy Visualization Part 3



Model Explainability: SHAP Analysis

Senior Decision-Makers

VP, Director, and C-suite roles are the strongest conversion drivers

- Incremental conversion gains
- Untapped opportunities
- Cost-effective wins

Quality & Regulatory Roles

Highest conversion probability among all functional areas

- Regulatory compliance need
- Quality systems; high conversion potential

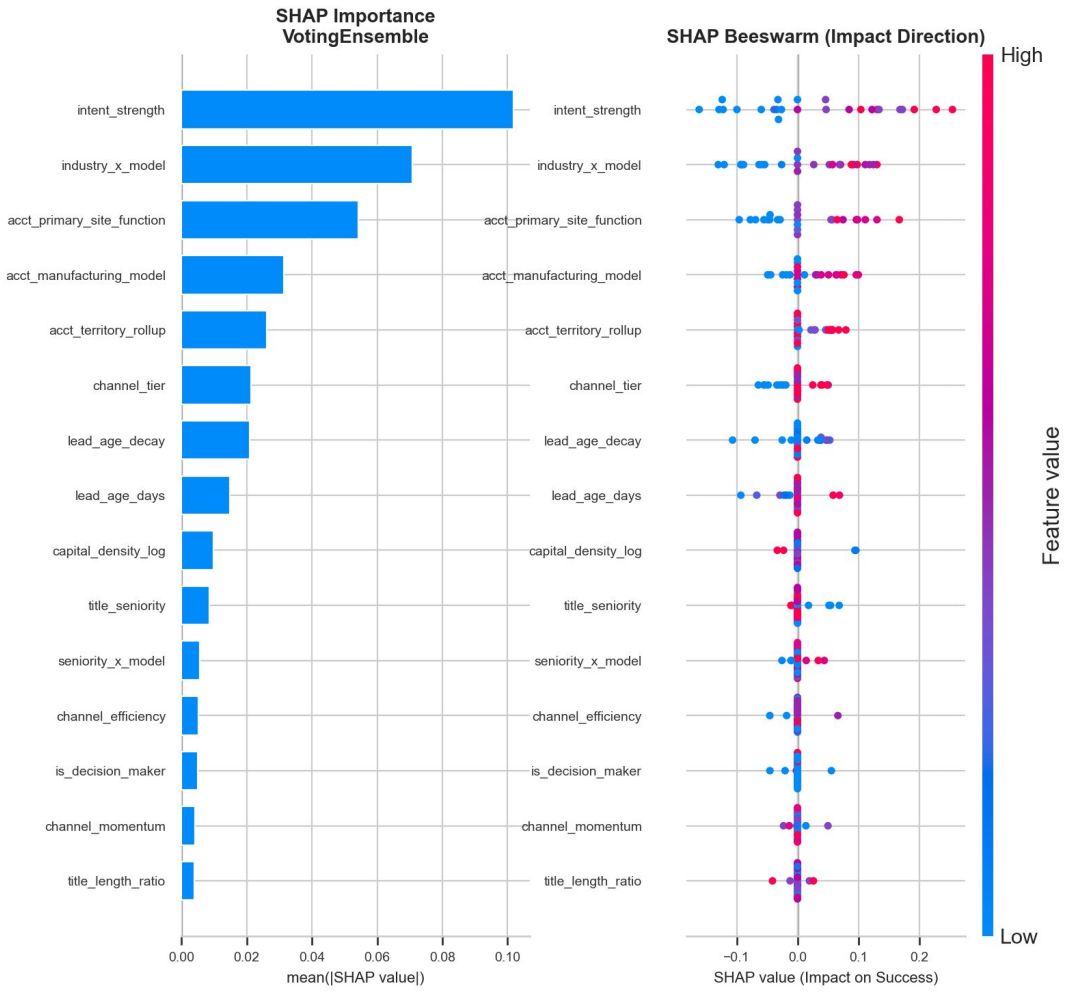
Premium Channels

Webinars, demos, and direct outreach significantly outperform low-value channels

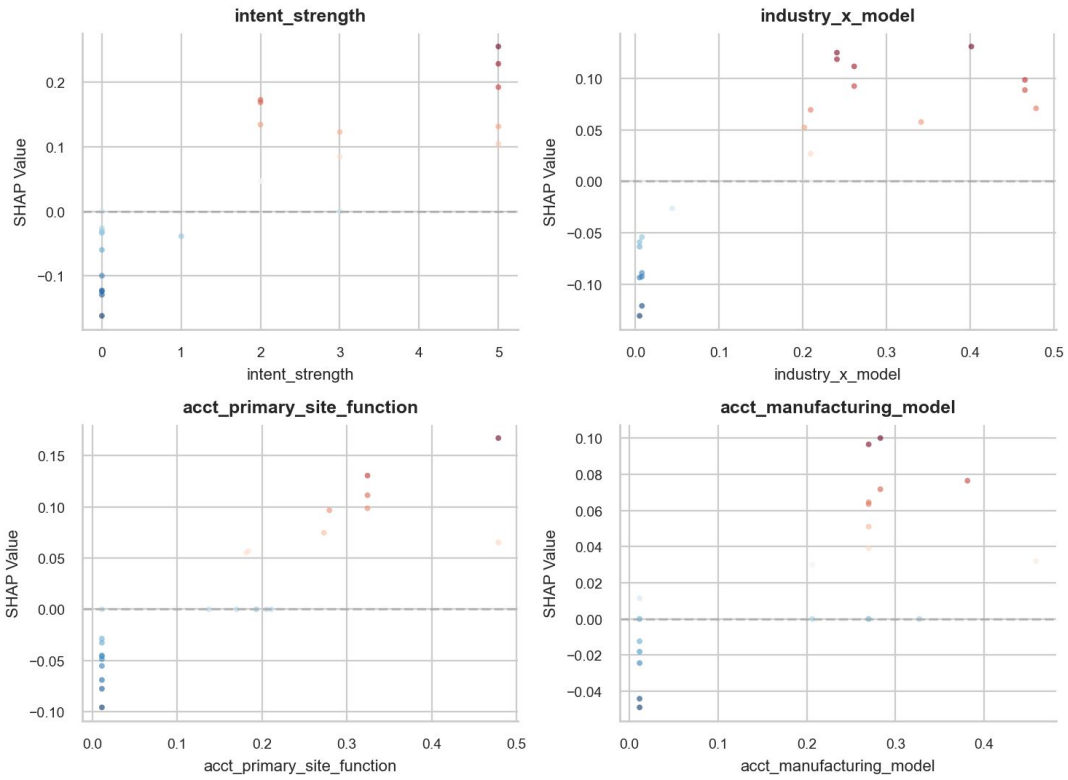
- Premium webinars
- Personalized demos
- Direct executive outreach

SHAP values reveal which features drive predictions, ensuring transparency and trust for sales adoption

SHAP Analysis Visualization



SHAP Analysis Visualization Part 2



Business Impact Quantification

Scoring

Model generates probability scores for all leads, ranking conversion likelihood.

Strategy

Ranked lead lists enable optimal SDR contact strategy and time allocation.

Selection

Profit maximization through data-driven lead selection and prioritization.

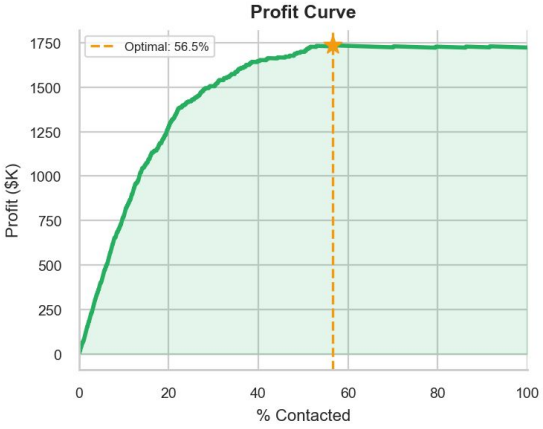
Allocation

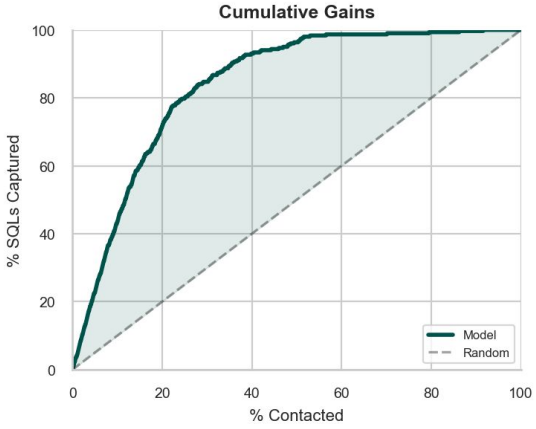
Resource allocation optimized based on conversion likelihood scores.

ROI

Measurable ROI improvement through targeted outreach to high-probability leads.

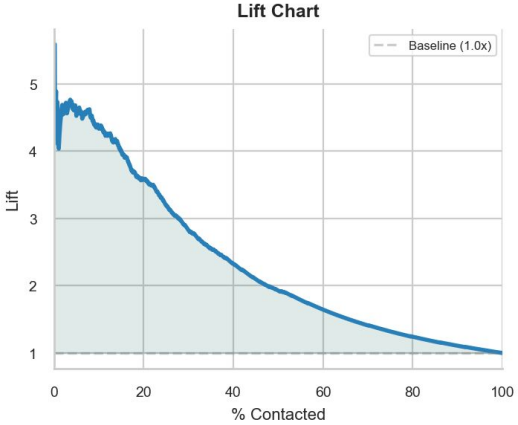
Business Impact Visualization

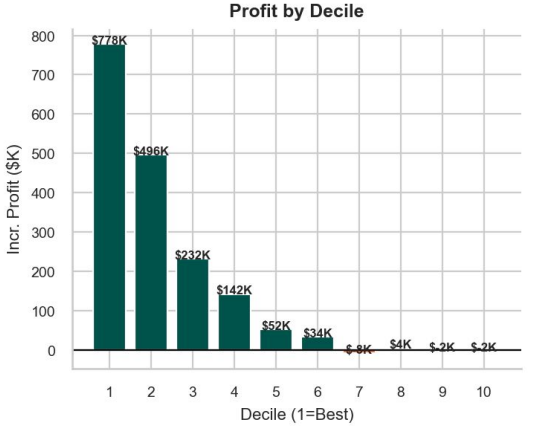




****Optimal Profit Configuration****

Metric	Value
Optimal Threshold	0.076
Leads to Contact	951
SQLs Captured	297
% Population	56.5%
SQL Capture Rate	98.7%
Net Profit	$1,734,450
Lift vs Random	1.7x





Final Model Summary KPIs

Metric	Value
Test AUC-ROC	0.9147
Average Precision	0.6993
Best Model	VotingEnsemble
Optimal Threshold	0.076
Max Profit (Test)	$1,734,450
Annual Lift	$150,600
Calls Required	951 (57%)
SQL Capture	99%
Top Decile Conv.	77.5%
Baseline Conv.	17.9%

Strategic Recommendations

Priority

Focus SDR time exclusively on top-ranked leads

- • High-probability leads
- • Maximize conversion, cut wasted outreach

Industry

Target Pharma and Biotech sectors

- • Regulatory compliance need
- • Quality systems; high conversion potential

Seniority

Engage senior decision-makers only

- • VP and Director levels
- • C-suite contacts
- • Budget authority holders

Channels

Invest in high-intent engagement channels

- • Premium webinars
- • Personalized demos
- • Direct executive outreach

Re-engagement

Re-engage high-scoring recycled leads

- • Incremental conversion gains
- • Untapped opportunities
- • Cost-effective wins

Conclusion and Key Takeaway

Actionable Sales Intelligence

Data-driven lead scoring transforms CRM data into actionable intelligence for sales teams

Closes the Conversion Gap

Improves Mx targeting precision and closes the 7% conversion gap with Qx

Direct Revenue Impact

Delivers measurable revenue impact by connecting SDRs with the right leads at the right time

Scalable Framework

Provides a scalable framework for continuous optimization and learning as new data emerges

THANK YOU

Any
Question?